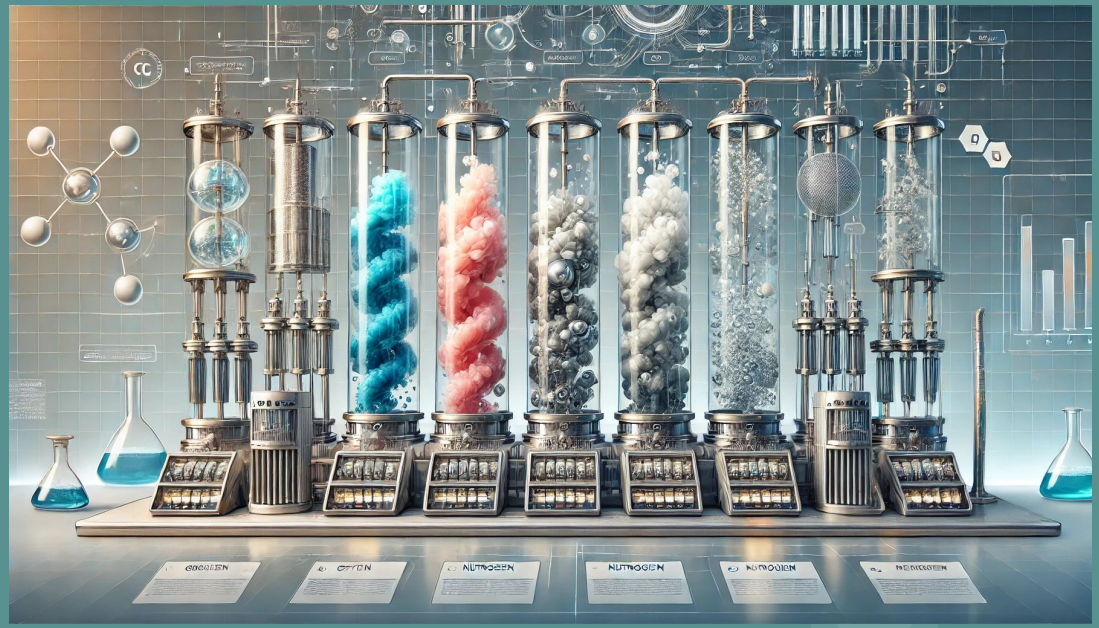
ST6: Gas separation



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Innhold

- Introduksjon
- Teori
- Eksperimentell
- Resultater
- Diskusjon

Introduksjon

Separering av oksygen- og nitrogengass

Membran-gasseparasjon

Forskjeller i gass permeabilitet

Teori - Bakgrunnsinfo

Selektiv barriere mellom to faser

Ikke-porøse polymermembraner

Nøkkelparametre

1. Selektivitet
2. Diffusjonskoeffisient
3. Sorpsjon

Teori - Likninger

Massebalanse:

$$L_f = L_o + V_p$$

“Stage cut” :

$$\theta = V_p / L_f$$

“Recovery”

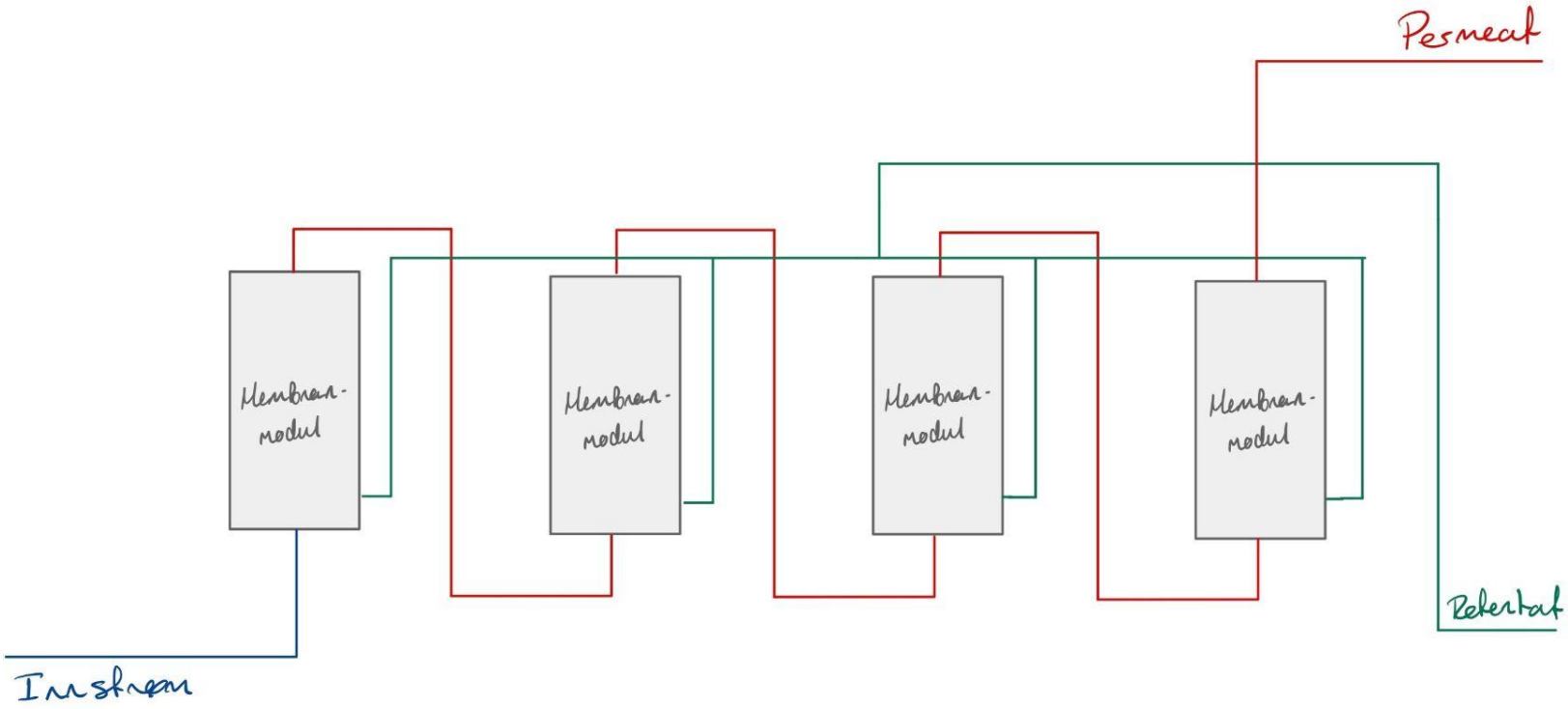
$$R_A = (X_{o,A} L_o)/(X_{f,A} L_f) \cdot 100\%$$

Molfraksjon:

$$X_{f,O2} = (L_o \cdot X_{o,O2} + V_p \cdot X_{o,O2}) / L_f$$

Trykkforholdet:

$$\varphi = p_f / p_p$$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Membran-
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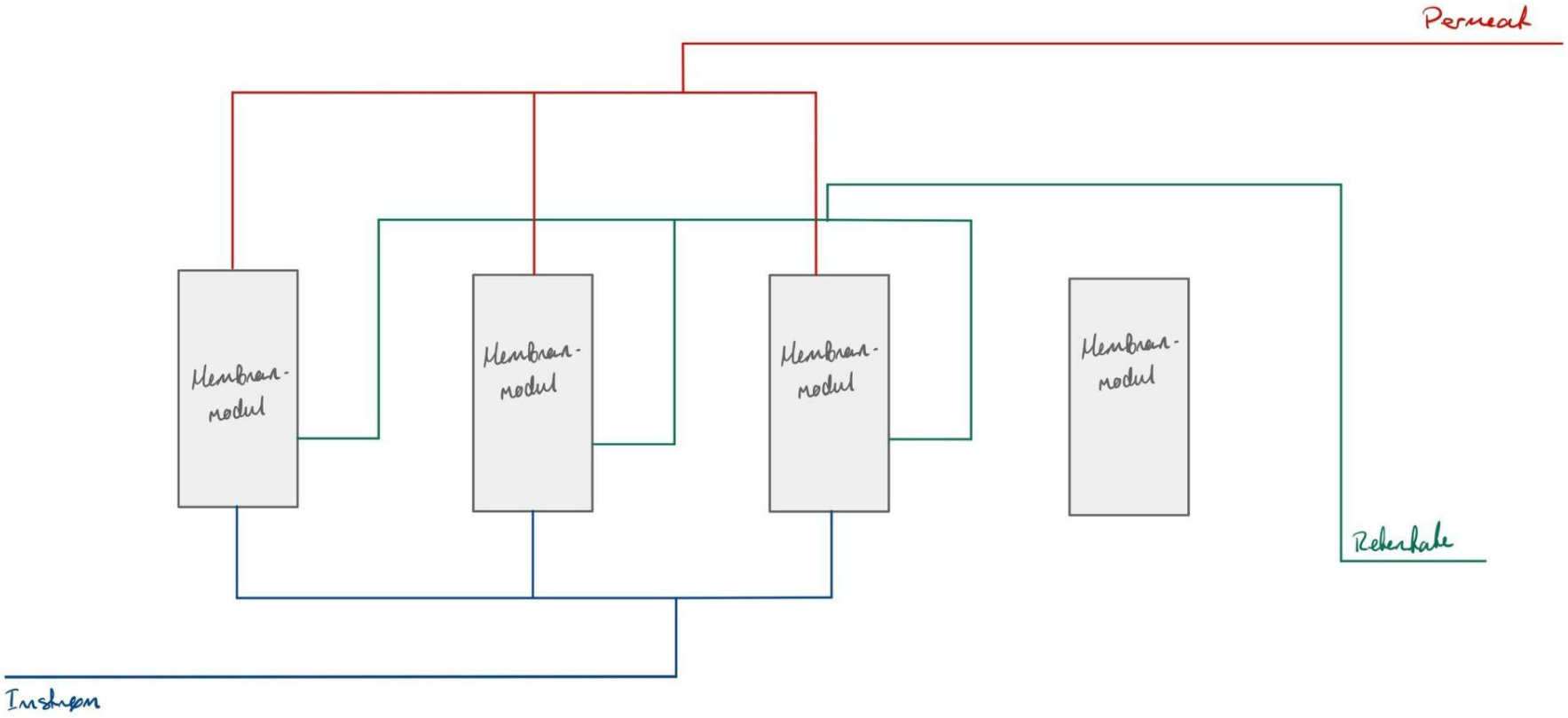
Retentat
Instrom


Permeat
Membran-
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Membran-
modul
Membran-
modul
Membran-
modul
Retentate
Instrom

Eksperimentell prosedyre

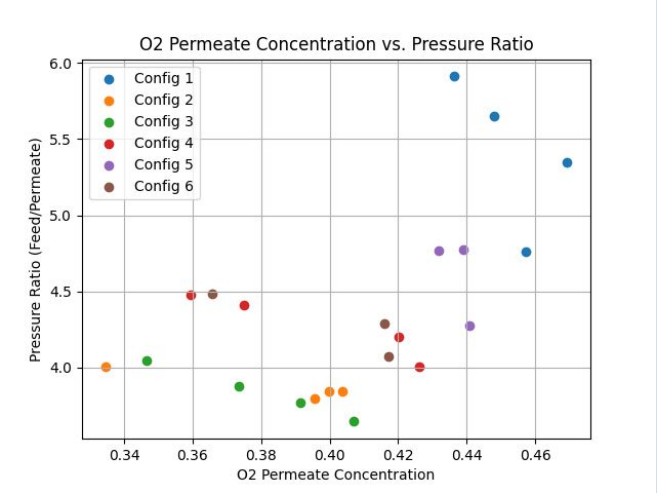
1. Lukk hovedmatingventilen
2. Åpne gassutløpsventilen og sett utløpstrykket til ca. 5 bar
3. Gjør nødvendige koblinger og sjekk for lekkasjer
4. Start programmet “run-py”
5. Sett ønsket strømningshastighet og bekreft
6. Åpne feed-ventilen sakte

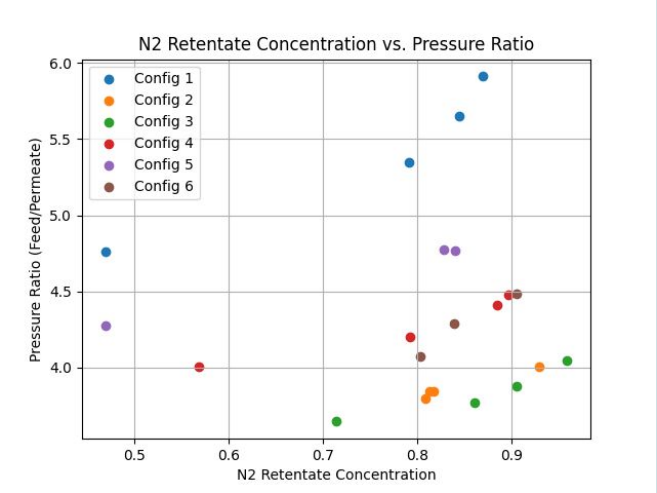
1. Vent til steady state og start prøvetaking
2. Stopp prøvetaking når kjøringen er ferdig
3. For å endre strømningshastighet, gjenta steg 4-8
4. Når eksperimentet er ferdig, stopp prøvetaking, lukk hovedventilen og lagre data
5. Sjekk at matingstrykket er nær 1 bar
6. Endre konfigurasjon og start fra punkt 2

Membranmoduler

1. Enkeltmembran
2. Fire membraner koblet i serie
3. Fire membraner koblet i parallell
4. Tre membraner koblet i parallell
5. Tre membraner i serie
6. Tre membraner i parallell

Resultater- Permeate konsentrasjon vs. pressure ratio

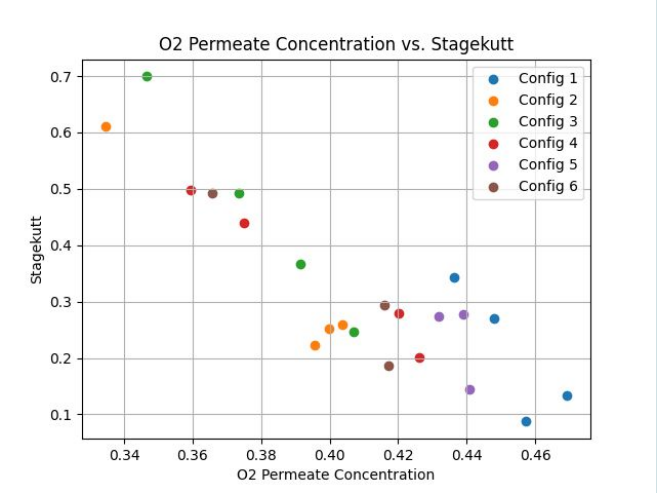




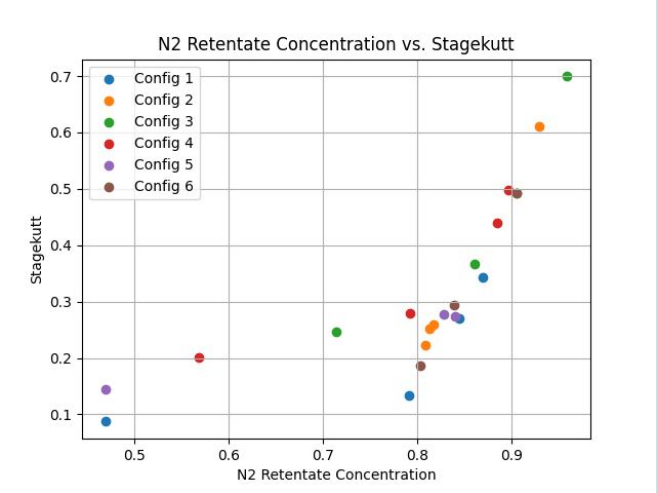
Recovery av o2 i permeat, og konsentrasjon av o2 i permeat - grafen viser generelt et negativt stigningstall for alle konfigurasjonene. Teorien sier at man bør få høyere konsentrasjon av oksygen i permeatet når recovery er høy.

Recovery av n2 i retentat, og konsentrasjon av n2 i i retentat - får generelt høy konsentrasjon uavhengig av recovery. Ønsker at konsentrasjonen skal stige med økt recovery, men får heller generelt et negativt stigningstall.

Resultater- Permeate konsentrasjon vs. stagekutt

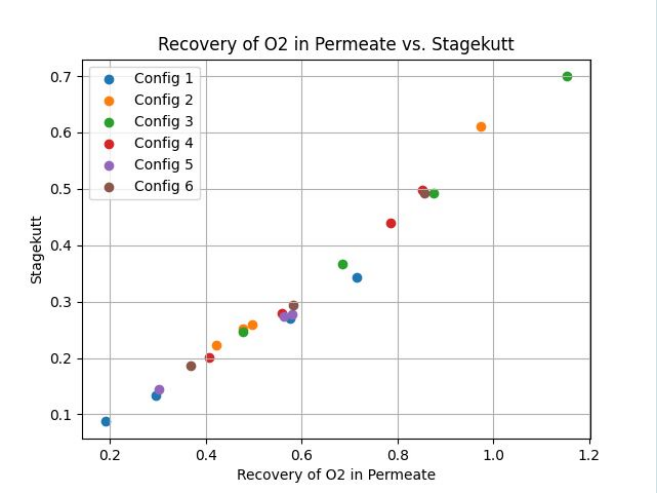


Recovery av o2i permeat, og stagekutt - vanskelig å skille konfigurasjonene, men trenden er at alle følger teorien, som tilsier at recovery i permeat øker med økt stagekutt.

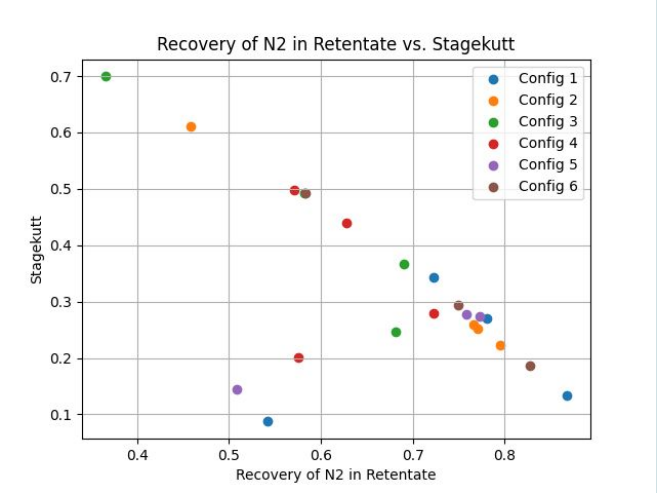


Recovery av o2i permeat, og stagekutt - vanskelig å skille konfigurasjonene, men trenden er at alle følger teorien, som tilsier at recovery i permeat øker med økt stagekutt.

Resultater- Recovery i permeate vs. stagekutt

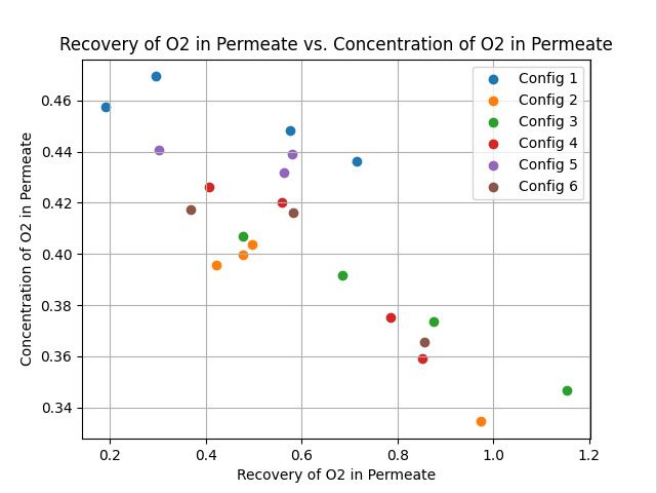


o2 permeat-konsentrasjon, og stagekutt - høy konsentrasjon når det er lav stagekutt. Det gir mening, ettersom lavt stagekutt tilsvarer lav selektivitet, som igjen vil gi høyere konsentrasjon.

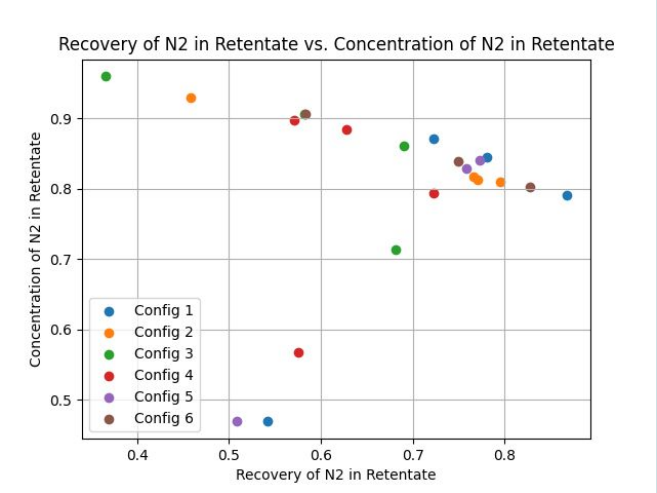


n2 retentat-konsentrasjon, og stagekutt - gir mening at grafen stort sett starter på litt under 0.8, ettersom det er 78% n2 i atmosfæren.

Resultater- Recovery i permeate vs. konsentrasjon



o2 permeat-konsentrasjon, og trykkrate - for høyere trykkrate, er trykket fra feeden høyere enn det fra permeatet. Det vil tvinge mere o2 over membranen, som vil gi høyere konsentrasjon av o2 i permeat-strømmen



n2 retentat-konsentrasjon, og trykkrate - følger samme trend som o2, for høyere konsentrasjon av o2 i permeat, gir det mening at konsentrasjonen av n2 i retentatet stiger.

```python
import numpy as np
import matplotlib.pyplot as plt

# Data for config_1
config_1_feed_P = [5.126, 5.541, 5.681, 4.513]
config_1_permeate_P = [0.958, 0.980, 0.961, 0.948]
config_1_retentate_P = [4.934, 5.478, 5.645, 4.039]
config_1_O2_permeate = [0.46939, 0.44812, 0.43637, 0.4574]
config_1_O2_retentate = [0.20896, 0.15545, 0.12977, 0.53114]
config_1_permeate_flow = [7.686, 9.289, 7.962, 6.920]
config_1_retentate_flow = [50.153, 25.147, 15.197, 72.32]

# Data for config_2
config_2_feed_P = [4.620, 4.458, 5.466, 4.628]
config_2_permeate_P = [1.202, 1.174, 1.364, 1.203]
config_2_retentate_P = [3.508, 3.508, 5.330, 3.846]
config_2_O2_permeate = [0.39975, 0.39577, 0.33448, 0.40367]
config_2_O2_retentate = [0.18686, 0.19080, 0.07002, 0.18278]
config_2_permeate_flow = [18.964, 18.060, 23.844, 19.065]
config_2_retentate_flow = [56.474, 62.755, 15.179, 54.524]

# Data for config_3
config_3_feed_P = [4.511, 5.043, 5.338, 5.565]
config_3_permeate_P = [1.237, 1.337, 1.377, 1.376]
config_3_retentate_P = [4.227, 4.901, 5.262, 5.528]
config_3_O2_permeate = [0.40703, 0.39148, 0.37346, 0.34664]
config_3_O2_retentate = [0.28612, 0.13877, 0.09430, 0.04061]
config_3_permeate_flow = [19.579, 23.135, 24.247, 24.136]
config_3_retentate_flow = [59.994, 39.887, 25.035, 10.373]
```

```python
# Data for config_4
config_4_feed_P = [4.488, 5.496, 4.934, 5.401]
config_4_permeate_P = [1.121, 1.227, 1.174, 1.225]
config_4_retentate_P = [4.180, 5.440, 4.755, 5.329]
config_4_O2_permeate = [0.42616, 0.35930, 0.42023, 0.37507]
config_4_O2_retentate = [0.43203, 0.10291, 0.20712, 0.11559]
config_4_permeate_flow = [16.108, 19.784, 18.057, 19.733]
config_4_retentate_flow = [64.309, 19.992, 46.502, 25.161]

# Data for config_5
config_5_feed_P = [4.396, 5.194, 5.220]
config_5_permeate_P = [1.028, 1.088, 1.094]
config_5_retentate_P = [3.989, 5.061, 5.084]
config_5_O2_permeate = [0.44084, 0.43897, 0.43194]
config_5_O2_retentate = [0.53116, 0.17131, 0.15927]
config_5_permeate_flow = [12.054, 14.805, 15.028]
config_5_retentate_flow = [71.531, 38.552, 39.923]

# Data for config_6
config_6_feed_P = [5.020, 5.467, 4.463]
config_6_permeate_P = [1.17, 1.219, 1.096]
config_6_retentate_P = [4.646, 5.344, 3.698]
config_6_O2_permeate = [0.41608, 0.36567, 0.41726]
config_6_O2_retentate = [0.16092, 0.09419, 0.19717]
config_6_permeate_flow = [17.912, 19.549, 15.117]
config_6_retentate_flow = [42.909, 20.214, 66.293]
```

```python
# Calculations
#-----------------------------------------------------------------------------------------------
# Mass balance
L_f_1 = [L_0 + V_p for L_0, V_p in zip(config_1_retentate_flow, config_1_permeate_flow)]
L_f_2 = [L_0 + V_p for L_0, V_p in zip(config_2_retentate_flow, config_2_permeate_flow)]
L_f_3 = [L_0 + V_p for L_0, V_p in zip(config_3_retentate_flow, config_3_permeate_flow)]
L_f_4 = [L_0 + V_p for L_0, V_p in zip(config_4_retentate_flow, config_4_permeate_flow)]
L_f_5 = [L_0 + V_p for L_0, V_p in zip(config_5_retentate_flow, config_5_permeate_flow)]
L_f_6 = [L_0 + V_p for L_0, V_p in zip(config_6_retentate_flow, config_6_permeate_flow)]

# Calculate pressure ratios
PR_1 = [feed / permeate for feed, permeate in zip(config_1_feed_P, config_1_permeate_P)]
PR_2 = [feed / permeate for feed, permeate in zip(config_2_feed_P, config_2_permeate_P)]
PR_3 = [feed / permeate for feed, permeate in zip(config_3_feed_P, config_3_permeate_P)]
PR_4 = [feed / permeate for feed, permeate in zip(config_4_feed_P, config_4_permeate_P)]
PR_5 = [feed / permeate for feed, permeate in zip(config_5_feed_P, config_5_permeate_P)]
PR_6 = [feed / permeate for feed, permeate in zip(config_6_feed_P, config_6_permeate_P)]

# Calculate Stagekutt
SK_1 = [Vp / Lf for Vp, Lf in zip(config_1_permeate_flow, L_f_1)]
SK_2 = [Vp / Lf for Vp, Lf in zip(config_2_permeate_flow, L_f_2)]
SK_3 = [Vp / Lf for Vp, Lf in zip(config_3_permeate_flow, L_f_3)]
SK_4 = [Vp / Lf for Vp, Lf in zip(config_4_permeate_flow, L_f_4)]
SK_5 = [Vp / Lf for Vp, Lf in zip(config_5_permeate_flow, L_f_5)]
SK_6 = [Vp / Lf for Vp, Lf in zip(config_6_permeate_flow, L_f_6)]

# Calculating N2 retentate for each configuration
config_1_N2_retentate = [1 - O2 for O2 in config_1_O2_retentate]
config_2_N2_retentate = [1 - O2 for O2 in config_2_O2_retentate]
```

```python
config_3_N2_retentate = [1 - O2 for O2 in config_3_O2_retentate]
config_4_N2_retentate = [1 - O2 for O2 in config_4_O2_retentate]
config_5_N2_retentate = [1 - O2 for O2 in config_5_O2_retentate]
config_6_N2_retentate = [1 - O2 for O2 in config_6_O2_retentate]

# Calculating N2 permeate for each configuration
config_1_N2_permeate = [1 - O2 for O2 in config_1_O2_permeate]
config_2_N2_permeate = [1 - O2 for O2 in config_2_O2_permeate]
config_3_N2_permeate = [1 - O2 for O2 in config_3_O2_permeate]
config_4_N2_permeate = [1 - O2 for O2 in config_4_O2_permeate]
config_5_N2_permeate = [1 - O2 for O2 in config_5_O2_permeate]
config_6_N2_permeate = [1 - O2 for O2 in config_6_O2_permeate]


# Feed fractions
Xf_O2_feed=0.21
Xf_N2_feed=0.79

# Recovery of O2 in permeate
R_O2_1 = [x_0 * V_0 / (x_f * L_f) for x_0, V_0, x_f, L_f in zip(config_1_O2_permeate, config_1_permeate_flow, [Xf_O2_feed] * len(config_1_O2_permeate), L_f_1)]
R_O2_2 = [x_0 * V_0 / (x_f * L_f) for x_0, V_0, x_f, L_f in zip(config_2_O2_permeate, config_2_permeate_flow, [Xf_O2_feed] * len(config_2_O2_permeate), L_f_2)]
R_O2_3 = [x_0 * V_0 / (x_f * L_f) for x_0, V_0, x_f, L_f in zip(config_3_O2_permeate, config_3_permeate_flow, [Xf_O2_feed] * len(config_3_O2_permeate), L_f_3)]
R_O2_4 = [x_0 * V_0 / (x_f * L_f) for x_0, V_0, x_f, L_f in zip(config_4_O2_permeate, config_4_permeate_flow, [Xf_O2_feed] * len(config_4_O2_permeate), L_f_4)]
R_O2_5 = [x_0 * V_0 / (x_f * L_f) for x_0, V_0, x_f, L_f in zip(config_5_O2_permeate, config_5_permeate_flow, [Xf_O2_feed] * len(config_5_O2_permeate), L_f_5)]
R_O2_6 = [x_0 * V_0 / (x_f * L_f) for x_0, V_0, x_f, L_f in zip(config_6_O2_permeate, config_6_permeate_flow, [Xf_O2_feed] * len(config_6_O2_permeate), L_f_6)]

# Recovery of N2 in retentate
R_N2_1 = [x_0 * L_0 / (x_f * L_f) for x_0, L_0, x_f, L_f in zip(config_1_N2_retentate, config_1_retentate_flow, [Xf_N2_feed] * len(config_1_N2_retentate), L_f_1)]
R_N2_2 = [x_0 * L_0 / (x_f * L_f) for x_0, L_0, x_f, L_f in zip(config_2_N2_retentate, config_2_retentate_flow, [Xf_N2_feed] * len(config_2_N2_retentate), L_f_2)]
```

```python
R_N2_3 = [x_0 * L_0 / (x_f * L_f) for x_0, L_0, x_f, L_f in zip(config_3_N2_retentate, config_3_retentate_flow, [Xf_N2_feed] * len(config_3_N2_
R_N2_4 = [x_0 * L_0 / (x_f * L_f) for x_0, L_0, x_f, L_f in zip(config_4_N2_retentate, config_4_retentate_flow, [Xf_N2_feed] * len(config_4_N2_retentate), L_f_4)]
R_N2_5 = [x_0 * L_0 / (x_f * L_f) for x_0, L_0, x_f, L_f in zip(config_5_N2_retentate, config_5_retentate_flow, [Xf_N2_feed] * len(config_5_N2_retentate), L_f_5)]
R_N2_6 = [x_0 * L_0 / (x_f * L_f) for x_0, L_0, x_f, L_f in zip(config_6_N2_retentate, config_6_retentate_flow, [Xf_N2_feed] * len(config_6_N2_retentate), L_f_6)]

#-----------------------------------------------------------------------------------------------
# Plots
#-----------------------------------------------------------------------------------------------
# Plotting consentartion of N2 retentate vs. pressure ratio
plt.scatter(config_1_N2_retentate, PR_1, label='Config 1')
plt.scatter(config_2_N2_retentate, PR_2, label='Config 2')
plt.scatter(config_3_N2_retentate, PR_3, label='Config 3')
plt.scatter(config_4_N2_retentate, PR_4, label='Config 4')
plt.scatter(config_5_N2_retentate, PR_5, label='Config 5')
plt.scatter(config_6_N2_retentate, PR_6, label='Config 6')

# Adding labels and title
plt.xlabel("N2 Retentate Concentration")
plt.ylabel("Pressure Ratio (Feed/Permeate)")
plt.title("N2 Retentate Concentration vs. Pressure Ratio")
plt.legend()
plt.grid(True)
plt.show()


# Plotting consentartion of O2 retentate vs. pressure ratio
plt.scatter(config_1_O2_permeate, PR_1, label='Config 1')
plt.scatter(config_2_O2_permeate, PR_2, label='Config 2')
plt.scatter(config_3_O2_permeate, PR_3, label='Config 3')
```

```python
plt.scatter(config_4_O2_permeate, PR_4, label='Config 4')
plt.scatter(config_5_O2_permeate, PR_5, label='Config 5')
plt.scatter(config_6_O2_permeate, PR_6, label='Config 6')

# Adding labels and title
plt.xlabel("O2 Permeate Concentration")
plt.ylabel("Pressure Ratio (Feed/Permeate)")
plt.title("O2 Permeate Concentration vs. Pressure Ratio")
plt.legend()
plt.grid(True)
plt.show()


# Plotting concentration of N2 retentate vs. stagekutt
plt.scatter(config_1_N2_retentate, SK_1, label='Config 1')
plt.scatter(config_2_N2_retentate, SK_2, label='Config 2')
plt.scatter(config_3_N2_retentate, SK_3, label='Config 3')
plt.scatter(config_4_N2_retentate, SK_4, label='Config 4')
plt.scatter(config_5_N2_retentate, SK_5, label='Config 5')
plt.scatter(config_6_N2_retentate, SK_6, label='Config 6')

# Adding labels and title
plt.xlabel("N2 Retentate Concentration")
plt.ylabel("Stagekutt")
plt.title("N2 Retentate Concentration vs. Stagekutt")
plt.legend()
plt.grid(True)
plt.show()
```

```python
# Plotting concentration of O2 permeate vs. stagekutt
plt.scatter(config_1_O2_permeate, SK_1, label='Config 1')
plt.scatter(config_2_O2_permeate, SK_2, label='Config 2')
plt.scatter(config_3_O2_permeate, SK_3, label='Config 3')
plt.scatter(config_4_O2_permeate, SK_4, label='Config 4')
plt.scatter(config_5_O2_permeate, SK_5, label='Config 5')
plt.scatter(config_6_O2_permeate, SK_6, label='Config 6')

# Adding labels and title
plt.xlabel("O2 Permeate Concentration")
plt.ylabel("Stagekutt")
plt.title("O2 Permeate Concentration vs. Stagekutt")
plt.legend()
plt.grid(True)
plt.show()


# Plotting of recovery of N2 in retentate vs. stagekutt
plt.scatter(R_N2_1, SK_1, label='Config 1')
plt.scatter(R_N2_2, SK_2, label='Config 2')
plt.scatter(R_N2_3, SK_3, label='Config 3')
plt.scatter(R_N2_4, SK_4, label='Config 4')
plt.scatter(R_N2_5, SK_5, label='Config 5')
plt.scatter(R_N2_6, SK_6, label='Config 6')

# Adding labels and title for N2 retentate plot
plt.xlabel("Recovery of N2 in Retentate")
plt.ylabel("Stagekutt")
plt.title("Recovery of N2 in Retentate vs. Stagekutt")
```

```python
plt.legend()
plt.grid(True)
plt.show()


# Plotting of recovery of O2 in permeate vs. stagekutt
plt.scatter(R_O2_1, SK_1, label='Config 1')
plt.scatter(R_O2_2, SK_2, label='Config 2')
plt.scatter(R_O2_3, SK_3, label='Config 3')
plt.scatter(R_O2_4, SK_4, label='Config 4')
plt.scatter(R_O2_5, SK_5, label='Config 5')
plt.scatter(R_O2_6, SK_6, label='Config 6')

# Adding labels and title for O2 permeate plot
plt.xlabel("Recovery of O2 in Permeate")
plt.ylabel("Stagekutt")
plt.title("Recovery of O2 in Permeate vs. Stagekutt")
plt.legend()
plt.grid(True)
plt.show()


# Plotting of recovery of N2 in retentate vs. concentration of N2 in retentate
plt.scatter(R_N2_1, config_1_N2_retentate, label='Config 1')
plt.scatter(R_N2_2, config_2_N2_retentate, label='Config 2')
plt.scatter(R_N2_3, config_3_N2_retentate, label='Config 3')
plt.scatter(R_N2_4, config_4_N2_retentate, label='Config 4')
plt.scatter(R_N2_5, config_5_N2_retentate, label='Config 5')
plt.scatter(R_N2_6, config_6_N2_retentate, label='Config 6')
```

```python
# Adding labels and title for N2 retentate plot
plt.xlabel("Recovery of N2 in Retentate")
plt.ylabel("Concentration of N2 in Retentate")
plt.title("Recovery of N2 in Retentate vs. Concentration of N2 in Retentate")
plt.legend()
plt.grid(True)
plt.show()


# Plotting of recovery of O2 in permeate vs. concentration of O2 in permeate
plt.scatter(R_O2_1, config_1_O2_permeate, label='Config 1')
plt.scatter(R_O2_2, config_2_O2_permeate, label='Config 2')
plt.scatter(R_O2_3, config_3_O2_permeate, label='Config 3')
plt.scatter(R_O2_4, config_4_O2_permeate, label='Config 4')
plt.scatter(R_O2_5, config_5_O2_permeate, label='Config 5')
plt.scatter(R_O2_6, config_6_O2_permeate, label='Config 6')

# Adding labels and title for O2 permeate plot
plt.xlabel("Recovery of O2 in Permeate")
plt.ylabel("Concentration of O2 in Permeate")
plt.title("Recovery of O2 in Permeate vs. Concentration of O2 in Permeate")
plt.legend()
plt.grid(True)
plt.show()
```

Konklusjon

Konfigurasjon 1, med en enkelt membran, gir høyest o2-konsentrasjon i permeatet. Dette er uforventet, ettersom flere membraner koblet i serie, teoretisk sett bør gi en høyere konsentrasjon. Det er også verdt å nevne at konfigurasjon 2 og 3, koblet i henholdsvis serie og parallel, gir veldig like resultater, noe som er uventet, ettersom teorien tilsier at seriekobling bør gi høyere o2-konsentrasjon.